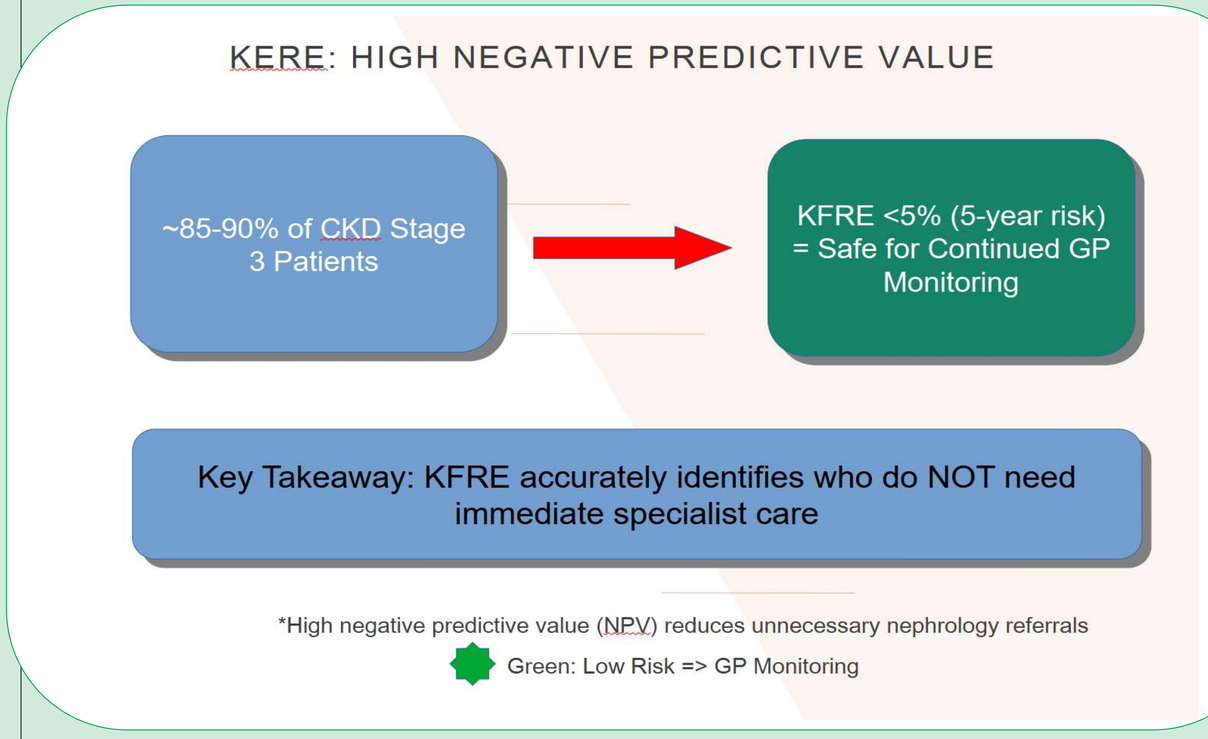
Higher Diploma in Computer Science, SETU

Developing the CareModel Hub Web Application for Implementing the Kidney Failure Risk Equation

Presenter: Peter Chuk

- *Chronic Kidney Disease (CKD), also referred to as Chronic Renal Disease(CRD), affects approximately 12% of the Irish population. Current detection rates remain around 40%, and while only ~3% of CRD patients progress to End Stage Renal Disease (ESRD), the absolute number of patients requiring secondary care services, including in-centre dialysis, continues to rise.*
- *The majority of CRD patients can be effectively managed within primary care settings, provided that high-risk individuals are identified early.*
- *The Kidney Failure Risk Equation (KFRE) predicts 5-year progression to ESRD using age, sex, eGFR, and ACR. Recommended by NICE NG203, KFRE enhances risk stratification, supports timely intervention, improves resource allocation, and reduces unnecessary secondary referrals.*



Regulatory & Governance

ISO 13485 (QMS) – Defines requirements for a structured Quality Management System for medical device software.

ISO 14971 (Risk) – Establishes a formal framework for identifying, analysing, and mitigating clinical and technical risks.

ISO 27001 (InfoSec) – Specifies controls for managing information security and protecting sensitive data.

MDR / IVDR – EU regulations governing the safety, performance, and clinical validation of medical and in-vitro diagnostic devices.

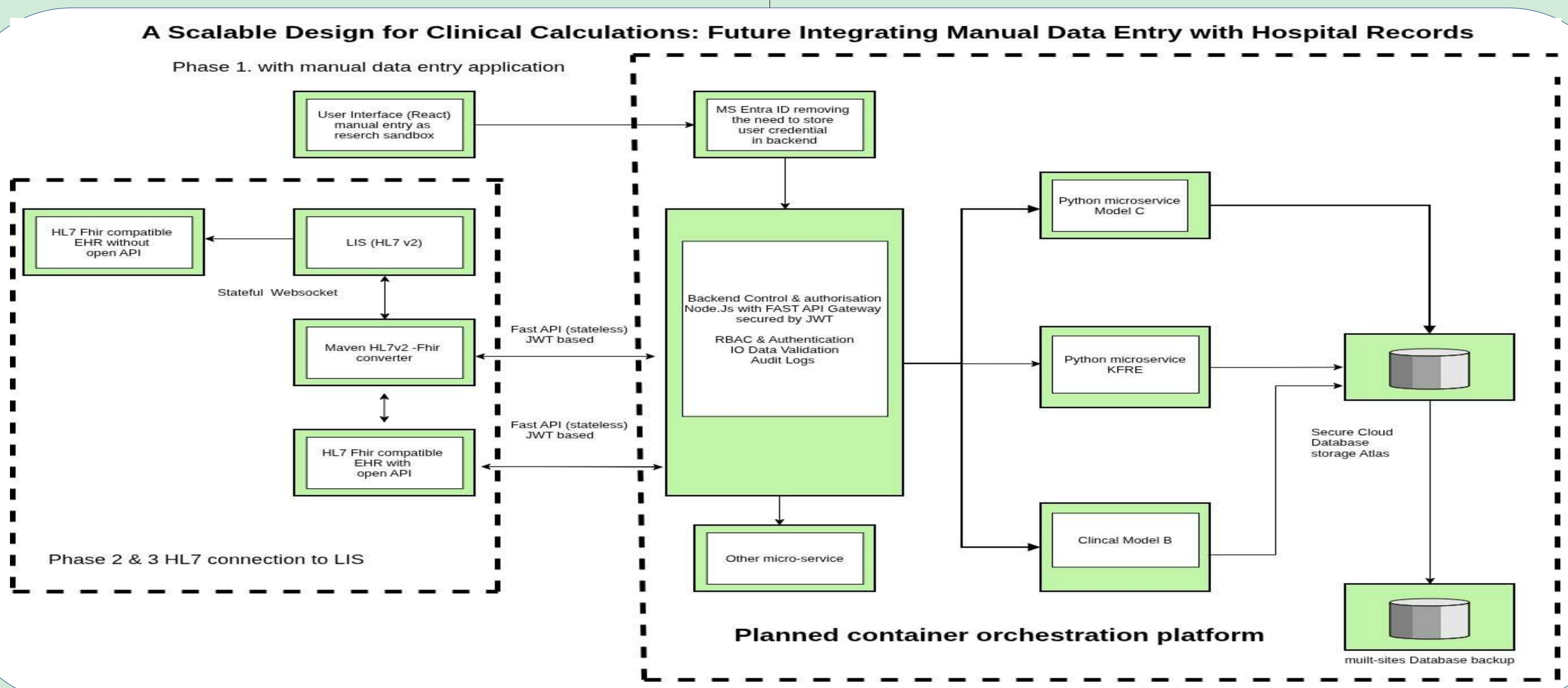
GDPR & NIS2 – European regulations ensuring lawful data protection and strengthened Cybersecurity resilience.

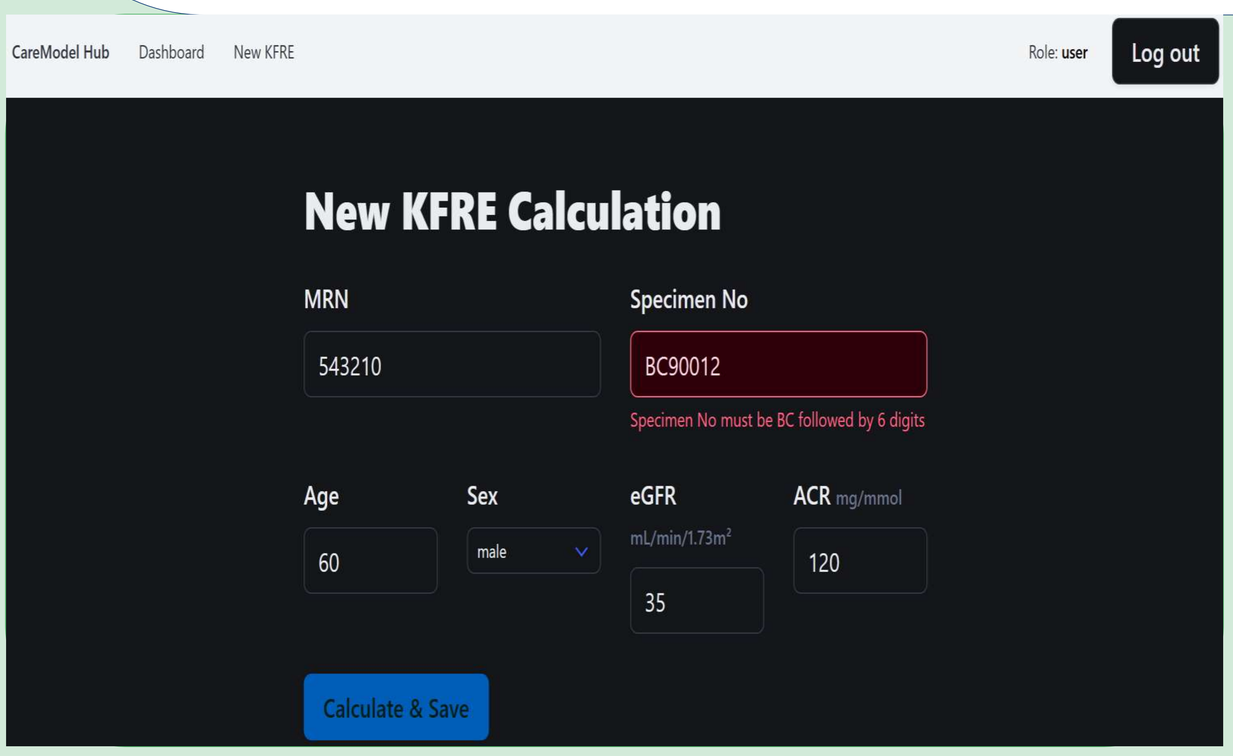
2. System Architecture Overview

Layered microservice architecture:

- React Frontend (UI Layer)
- Node.js (Hapi + TypeScript) Backend
- FastAPI Python Microservice (KFRE Engine)
- MongoDB Atlas Persistence

Design principles: modularity, security, fault isolation, scalability.





Scan QR code: https://pccork.github.io/caremodelhub/

email	Password	Role
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